

RAID 0+1: Doing it right

Getting the best of both worlds: Security and Performance

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One way to safeguard against data loss from hard drive failures is to setup a RAID array in your machine. There are different RAID levels one can opt for and each have their own benefits. RAID 0 offers superior performance as data is written across two physical hard drives, so while write speeds get a boost, read speeds go through the roof. RAID 1 offers data security as your entire data is mirrored completely so if one drive fails, you still have all your data safe. Then comes RAID 1+0 or RAID 10 as it is more commonly known. Here, we have the best of both worlds with superior performance and data security and we are going to help you setup your own RAID array.

Checklist:

1. Your motherboard should support RAID 10 with at least four connectors or you'll have to go in for a dedicated RAID card. They offer much more features (check the box).
2. HDDs of same make (Considering RAID and available ports) For RAID

10 you'll need 4 HDDs of the same manufacturer, make and capacity. This is most economical for a personal RAID 10 array. As the next possible setup requires eight HDDs.

3. Ensure your power supply can handle the extra drives and have extra connectors.

You'll end up with exactly half the total capacity of the drives in RAID. So if you are using 4 x 1 TB drives then you'll have 2 TB in your RAID array. With this in mind, let us begin.

Step 1 - Connect your drives

Your drives need to be connected to the right ports for this to work. Check your motherboard manual for instructions regarding which ports you'll need to connect your 4 HDDs onto. Attach the power connectors and boot-up.

Step 2 - Back up your data elsewhere This is important as your data on the 4 drives will be wiped completely when the RAID is setup.

Step 3 - Boot up

As soon as your computer boots up, enter the BIOS menu and navigate to Storage Configuration option. We setup our system on a motherboard with AMI BIOS, so the option was on the Main window

DEDICATED RAID CARDS

When your motherboard doesn't support RAID but you still wish to setup an array you need to go get yourself a dedicated RAID card. LSI, Adaptec, Areca, etc. and well known companies that offer personal and enterprise RAID solutions. With a dedicated RAID card, you can increase your read/write performance by up to 50% and reduces your CPU usage by more than 50%. It's a double whammy. Also these cards have a huge cache so one needs to have a UPS or else data loss due to power cuts will be huge. Not to mention the extra features like SSD caching which are part of the package.

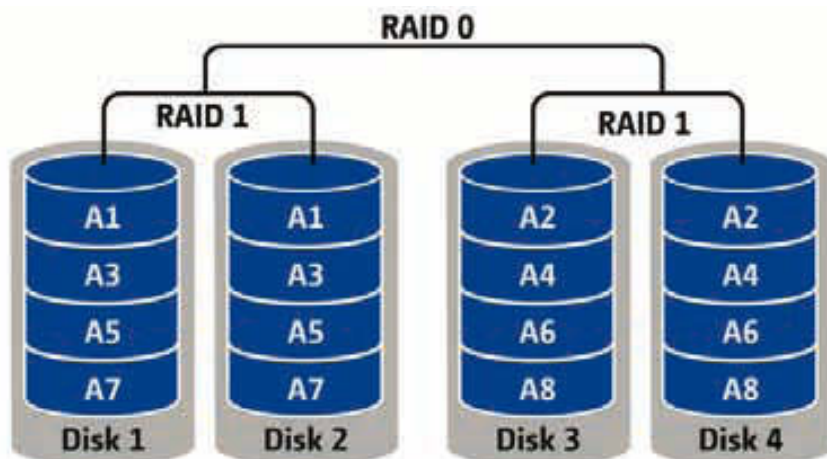
of the BIOS. Scroll down to "Configure SATA as" and select the "RAID" option. Save the configuration and reboot. Be ready to press "Ctrl+I" when prompted.

Step 4 - RAID setup

When you press "Ctrl+I" at the prompt, you'll be taken into the Intel Rapid Storage Technology/Matrix Storage Manager window. For AMD machines, the option will be "Ctrl+F". Here, name your configuration anything then select "RAID Level" as 10. Under "Disks", select the 4 drives you have connected to your machine and press enter. Let "Strip Size" remain 128 KB. Move on to Create volume and press enter and press Y to confirm. If your drives are empty, the configuration will take no time at all. Otherwise it will format each HDD individually and that takes a lot of time.

Step 5 - Boot into your OS.

Your RAID has been configured but you won't be able to access the array for any use yet. Under Windows, run "compmgmt.msc". Now under Storage > Disk Management you'll see your array as unallocated. Right click on the drive and select "New Simple Volume". Feed in the size of the partition and the label you wish to assign to the drive and then you'll have your partition ready for use. That's it. 



RAID 10 is equal to two RAID 1 arrays in RAID 0